

Snakebites

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- Venomous snakes in the United States include pit vipers (rattlesnakes, copperheads, and cottonmouths) and coral snakes.
- Severe venom injection (envenomation) can cause damage to the bitten extremity, bleeding, and vital organ damage.
- Venom antidote is given for serious bites.

(See also [Introduction to Bites and Stings](#).)

Bites from nonpoisonous snakes rarely cause any serious problems. There are many species of venomous snakes native to the United States. The venomous snakes include pit vipers (rattlesnakes, copperheads, and cottonmouths) and coral snakes. Animals that inject harmful toxins by biting or stinging are called venomous, while the term poisonous refers to exposure to toxins by eating the animal.

Of the roughly 45,000 snakebites that occur in the United States each year, fewer than 8000 are from venomous snakes, and about 5 people die. Fatal snakebites are much more common outside the United States.

In about 25% of all pit viper bites, venom is not injected. Most deaths occur in children, older adults, and people who are untreated or treated too late or inappropriately. Rattlesnakes account for the majority of snakebites and almost all snakebite-related deaths in the United States. Copperheads and, to a lesser extent, cottonmouths account for most other venomous snakebites. Coral snakebites and bites from imported snakes are much less common.

The venom of rattlesnakes and other pit vipers damages tissue around the bite. Venom may cause changes in blood cells, prevent blood from clotting, and damage blood vessels, causing them to leak. These changes can lead to internal bleeding and to heart, respiratory, and kidney failure.



Timber Rattlesnake

CDC/Edward J. Wozniak, DVM, PhD



Southern Copperhead

CDC/James Gathany



Eastern Cottonmouth

CDC/James Gathany



Coral Snake

Photo courtesy of Mike Cardwell (Victorville, CA).

The venom of coral snakes affects nervous system activity but causes little damage to tissue around the bite. Most bites occur on the hand or foot.

Did You Know...

- Snakebites can be terrifying but rarely cause deaths in the United States.

Symptoms of Snakebites

The symptoms of snake venom poisoning vary widely, depending on the following:

- The size and species of snake
- The amount and toxicity of the venom injected (related to the size and species of snake)
- The bite's location (the farther away from the head and trunk, the less dangerous)
- The person's age (very old and very young people are at higher risk)
- The person's underlying medical problems

Pit vipers

Bites by most pit vipers rapidly cause pain. Not all bites inject venom, but if the wound oozes, venom was probably injected. Redness and swelling usually follow within 30 to 60 minutes and can affect the entire leg or arm within several hours. Moderate or severe pit viper poisoning commonly causes bruising of the skin 3 to 6 hours after the bite. The skin around the bite appears tight and discolored. Blisters, often filled with

blood, may form in the bite area. Without treatment, tissue around the bite may be destroyed.

Other symptoms include fever, chills, general weakness, faintness, sweating, anxiety, confusion, nausea, vomiting, and diarrhea. Some of these symptoms may be caused by terror rather than venom. The gums may bleed, and blood may appear in the person's vomit, stools, and urine. Breathing difficulties can occur, particularly after Mojave rattlesnake bites. Some people may have chest pain. Hours later, people may have a headache, blurred vision, drooping eyelids, and a dry mouth.

Snakebite Wound



A blood-filled blister caused by a cottonmouth (water moccasin) bite.

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People bitten by a rattlesnake may experience tingling and numbness in the fingers or toes or around the mouth and a metallic or rubbery taste in the mouth.

Coral snakes

Coral snakebites usually cause little or no immediate pain and swelling. More severe symptoms may take several hours to develop. The area around the bite may tingle, and nearby muscles may become weak. Muscle incoordination and severe general weakness may follow. Other symptoms may include double vision, blurred vision, confusion, drowsiness, increased saliva production, and speech and swallowing difficulties. Breathing problems, which may be extreme, may develop.

Is That a Pit Viper?

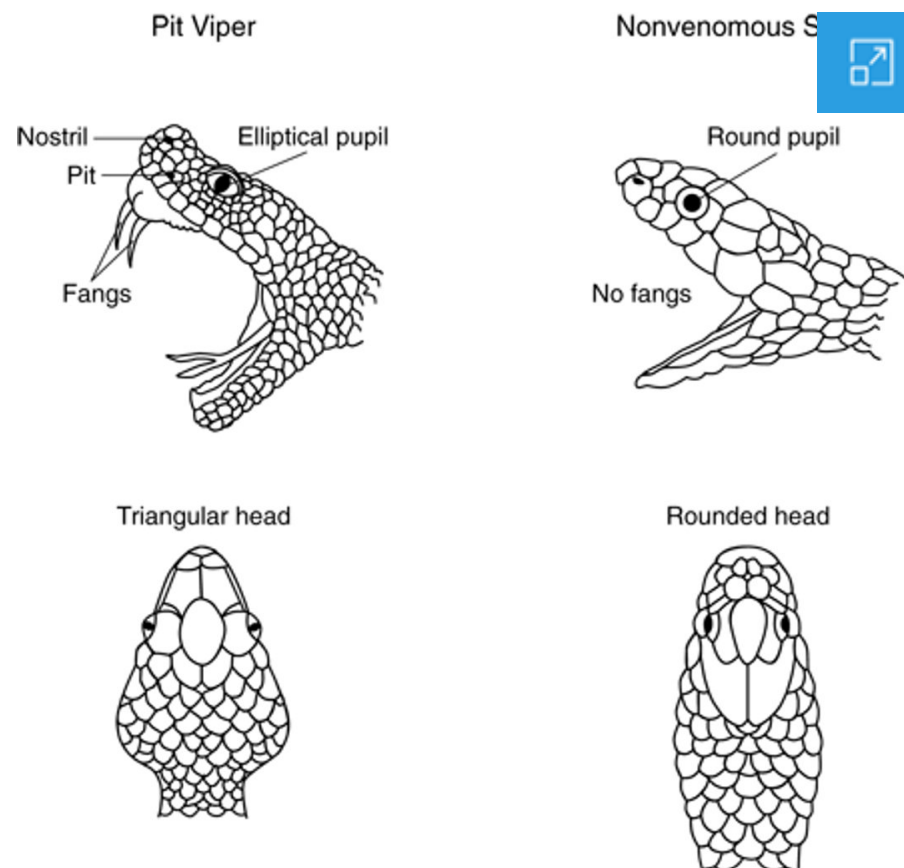
Pit vipers have certain features that can help distinguish them from nonvenomous snakes:

- Vertical slitlike pupils
- Pits between the eyes and nose
- Retractable fangs
- Rows of single scales across the underside of the tail
- Triangular heads (like an arrowhead)

Nonvenomous snakes tend to have the following:

- Rounded heads
- Round pupils
- No pits
- No fangs
- Rows of double scales across the underside of the tail

If people see a snake with no fangs, they should not assume it is nonvenomous because the fangs may be retracted.



Diagnosis of Snakebites

- A doctor's evaluation
- Hospitalization for observation for bites with envenomation

Emergency medical personnel must try to determine whether the snake was venomous, what species it was, and whether venom was injected.

The bite marks sometimes suggest whether the snake was venomous. The fangs of a venomous snake usually produce 1 or 2 large punctures, whereas the teeth of nonvenomous snakes usually leave multiple small rows of scratches. Without a detailed description of the snake, doctors may have difficulty determining the particular species that caused the bite.

Envenomation is recognized by the development of characteristic symptoms. People who are bitten by a venomous snake are generally kept in the hospital for observation for 6 to 8 hours to see if any symptoms develop. Doctors do various tests to assess the effects of the venom.

Treatment of Snakebites

- Bitten arm or leg is positioned at or above heart level and immobilized and constricting clothing and jewelry are removed
- Antivenom

First aid can be helpful before medical help arrives. People bitten by a poisonous snake should be moved beyond the snake's striking distance, kept as calm and still as possible, and taken to the nearest medical facility immediately.

The bitten limb should be loosely immobilized and kept positioned at about heart level or above. Rings, watches, and tight clothing should be removed from the area of the bite. Alcohol and caffeine should be avoided. Tourniquets, ice packs, and cutting the bite open are not recommended because they are potentially harmful. Applying suction to remove venom does not work.

If no venom was injected, treatment is the same as for any [puncture wound](#).

Venom antidote (antivenom) is the most important part of treatment if venom was injected and symptoms indicate a serious bite. Antivenom contains antibodies that neutralize venom's toxic effects. It is important to elevate the extremity as soon as the

antivenom is started to minimize local swelling. The effectiveness of antivenom depends on how soon it is given. Antivenom is more effective when it is given within a short time after a snakebite. It is given intravenously.

In the United States, antivenom is available for all native poisonous snakes.

Intensive care unit treatment is required for people with severe envenomation. People are monitored closely, and the complications of envenomation are treated. People with low blood pressure are given fluids intravenously. If problems with blood clotting develop, additional antivenom, fresh frozen plasma, concentrated clotting factors (cryoprecipitate), or platelet transfusions may be needed.

Prognosis depends on the person's age and overall health and on the location and venom content of the bite. Almost everyone bitten by a venomous snake survives if treated early with appropriate amounts of antivenom.

Serum sickness

Pit viper antivenom is made from the serum of sheep or horses that have been immunized with snake venom. Giving people foreign proteins such as antibodies from sheep or horse serum sometimes triggers an immune reaction called serum sickness.

Serum sickness causes fever, rash, and joint pains about 1 to 3 weeks after receiving the medication. Sometimes kidney damage occurs. Serum sickness occurs in about 1 in 6 people.

Doctors treat serum sickness with antihistamines, such as diphenhydramine, and corticosteroids.

Drug Information for the Topic



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